Assessment Method #2 (Direct / Summative):
Senior Written Exam
Sample Description: Spring 2015 Senior Written Examination and rubrics.

(See next 4 pages)

BSChE Senior Written Examination- Spring 2015

Scoring Rubrics Mapped to Student Learning Outcomes and Program Educational Objectives

Professionalism and Broader Knowledge

Question #	**Performance Indicator (PI)**	**SLO**	**(1)** Unsatisfactory	**(2)** Developing	**(3)** Satisfactory
1	1 Background research on career options and plans	**7**	No significant research. Unfamiliar with application deadlines, interview procedures and employer policies.	Has checked some information on specific universities or companies on the internet. Actively seeking job interviews or knows relevant deadlines for graduate schools.	Can quote employer policies (e.g., career tracks, promotions, sponsorship of advanced degrees). Has submitted applications, has job interviews scheduled, or already accepted for job or graduate school.
2a	2. Ethical and societal ramifications of technology	**4**	Unable to think of an example or cannot identify ethical ramifications of supplied technological topic.	Can independently come up with an example or identify an ethical ramification of supplied example.	Clearly explains ethical impact or social issues related to technological topic.
2c	3. Approach to ethical dilemmas	**4**	Unable to recognize ethical dilemma in hypothetical scenario. Purely ad hoc or “gut feel” approach toward ethical decisions. Does not know where or to whom to turn for guidance. Confounds fact-finding with ethical consideration.	Recognizes ethical dilemma in hypothetical scenario (or own experience). Logical considerations toward resolution. Can formulate steps to seek guidance.	Articulates resolution of dilemma as outcome of systematic thought process or ethical framework. Familiar with channels to air ethical concerns and/or cognizant officers.
2d	4. AIChE Code of Ethics	**4**	Unfamiliar with existence or content of code.	Knows where to find code and is familiar with some provisions.	Has read code more than once and can quote main provisions.
3	5. Multidisciplinary teamwork	**5**	Cannot name any collaborating engineering fields or explain context of collaboration.	Names one other type of engineer and articulates some aspects of collaboration.	Names two other types of engineer and clearly articulates example or specific contexts in which his/her work could intersect.

Technical Assessment

Question #	**Performance Indicator (PI)**	**SLO**	**(1)** Unsatisfactory	**(2)** Developing	**(3)** Satisfactory
4	6. Description of numerical method and formulation of algorithm	**1,2**	Does not know how to proceed and/or unable to get started.	Describes most of numerical method but shows some gaps in understanding or is cannot formulate algorithm.	Describes numerical method correctly and successfully formulates algorithm.
4	7.Proficiency with Excel or MATLAB	**1**	Unfamiliar with Excel, MATLAB (and any other math package). Unable to get started or needs extensive consultation of on-line manual to get started. Incomplete or incorrect solution.	Able to do significant portion or most of problem with some quick consultation of on-line help.	Fluent in finishing problem completely with little or no consultation of online help.
5	8.Identification of system, surroundings and applicable flow streams	**1,6**	Does not mention system or surroundings or falters seriously in implementation.	Provides many details of system, surroundings and flow streams correctly	Complete and correct formulation of engineering system.
5	9.Effective problem solving skills	**1**	Unable to translate physical ideas into mathematical form. Ineffective solution.	Formulates mathematical problem substantially and makes substantial progress toward correct solution	Correct mathematical formulation and completion of correct solution, including units

RESULTS BY PERFORMANCE INDICATOR:									
Rubric item (PI)	1	2	3	4	5	6	7	8	9
Mapped SLO	7	4	4	4	5	1,2	1	1,6	1
Average	2.76	2.54	2.57	2.13	2.63	2.57	2.59	2.37	2.41
Std Dev	0.42	0.72	0.54	0.88	0.53	0.62	0.58	0.68	0.69

RESULTS BY STUDENT OUTCOME ALL ASSESSMENTS						
	1	2	4	5	6	7
Average Score	2.48	2.57	2.41	2.63	2.37	2.76

BSChE Graduate Written Examination – Spring 2015
Written examination conducted with graduating seniors in Chemical Engineering
Name:

Question 1

Career plans. (Scoring rubric was designed to assess potential for life-long learning.)

Where do you see yourself in 5-10 years, and what steps will you take to get there?

How have you researched your career plans and options (companies, graduate schools, PE certification, etc.)?

Does your prospective employer sponsor master's degree or MBA? Under which terms?

Question 2

Professional ethics and ethical ramifications of technology.

Can you name a recent news item or significant piece of legislation that impacts on technology or engineering, and comment on its ethical ramifications?

Have you experienced an ethical dilemma in your educational or work (e.g., co-op) experience?

How would you approach an ethical dilemma? Where or to whom would you turn for guidance? Can ethical problems be approached in terms of a systematic methodology, or are you guided more by gut reaction, personal experience, upbringing, etc.?

Are you familiar with the AIChE Code of Ethics? Can you name specific provisions in it?

Which courses at UIC have contributed most to your views on professional ethics? How?

Question 3

Working in multidisciplinary teams

Can you name two other types of engineers that you might find yourself working with?

In what context or application would you be collaborating with them?

Question 4

Numerical methods and computational proficiency (Excel, MATLAB or Maple): choose one problem.

(a) Generate a graph of the solution of the following ODE initial value problem.

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1}{\tau}\Big[x_{\mathrm{in}}(t) - x(t)\Big] - kx(t)^2, \qquad x(0) = 2$$

$$x_{\mathrm{in}}(t) = 1 + \varepsilon \sin(\omega t)$$

with the constants $\tau = 3.4$, $k = 1.2$, $\varepsilon = 0.1$ and $\omega = 5$. Can you think of a chemical engineering problem (or, at least, a course) in which such an ODE would arise?

(b) Find a numerical value for the following definite integral.

$$\int_1^7 \frac{\mathrm{d}x}{1 + \cos^3 x + \ln x}$$

Question 5

Engineering estimate.

The waterwheel shown in the following three pictures was observed to make one complete revolution every 15 seconds. Estimate the power output (in units of horsepower)?